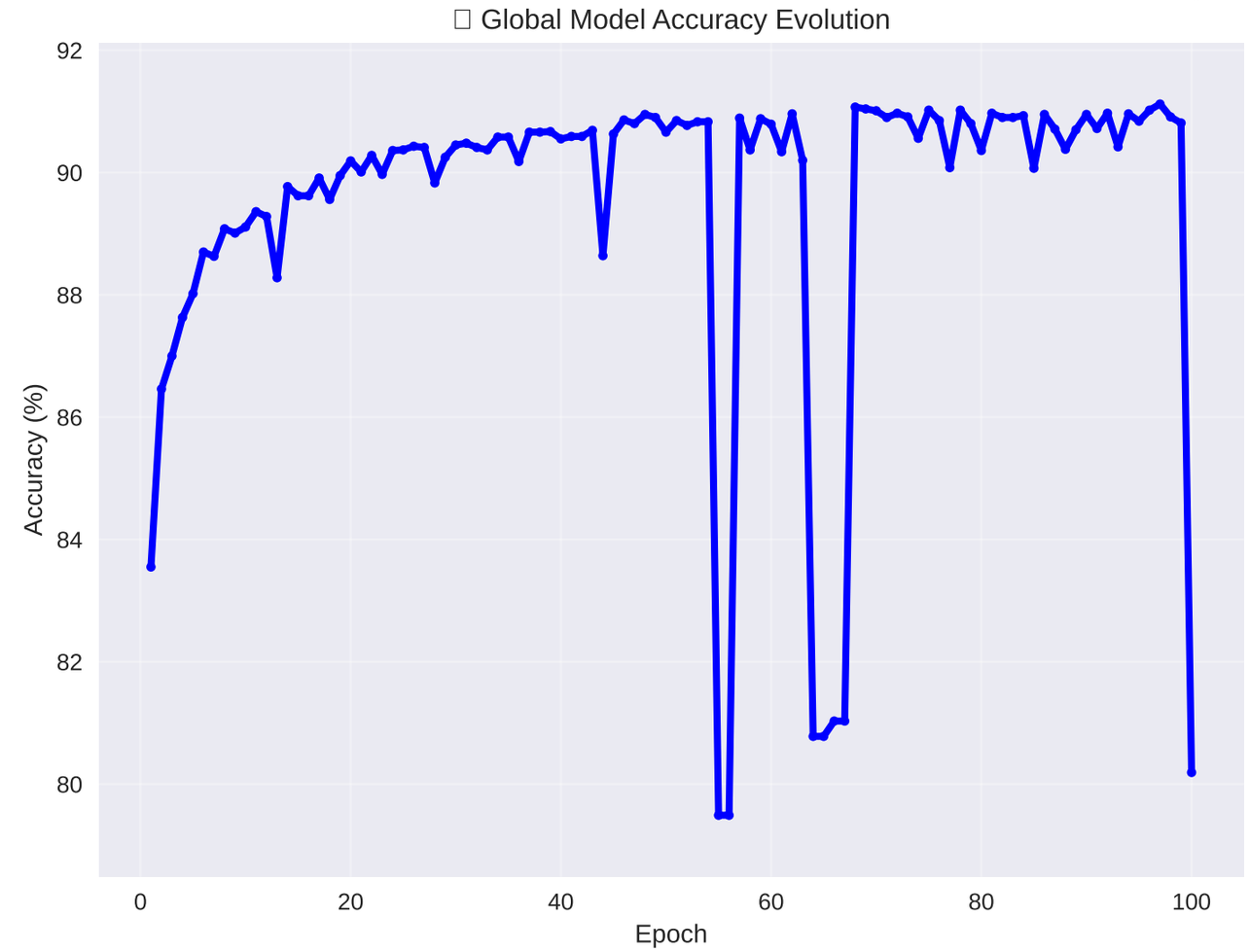
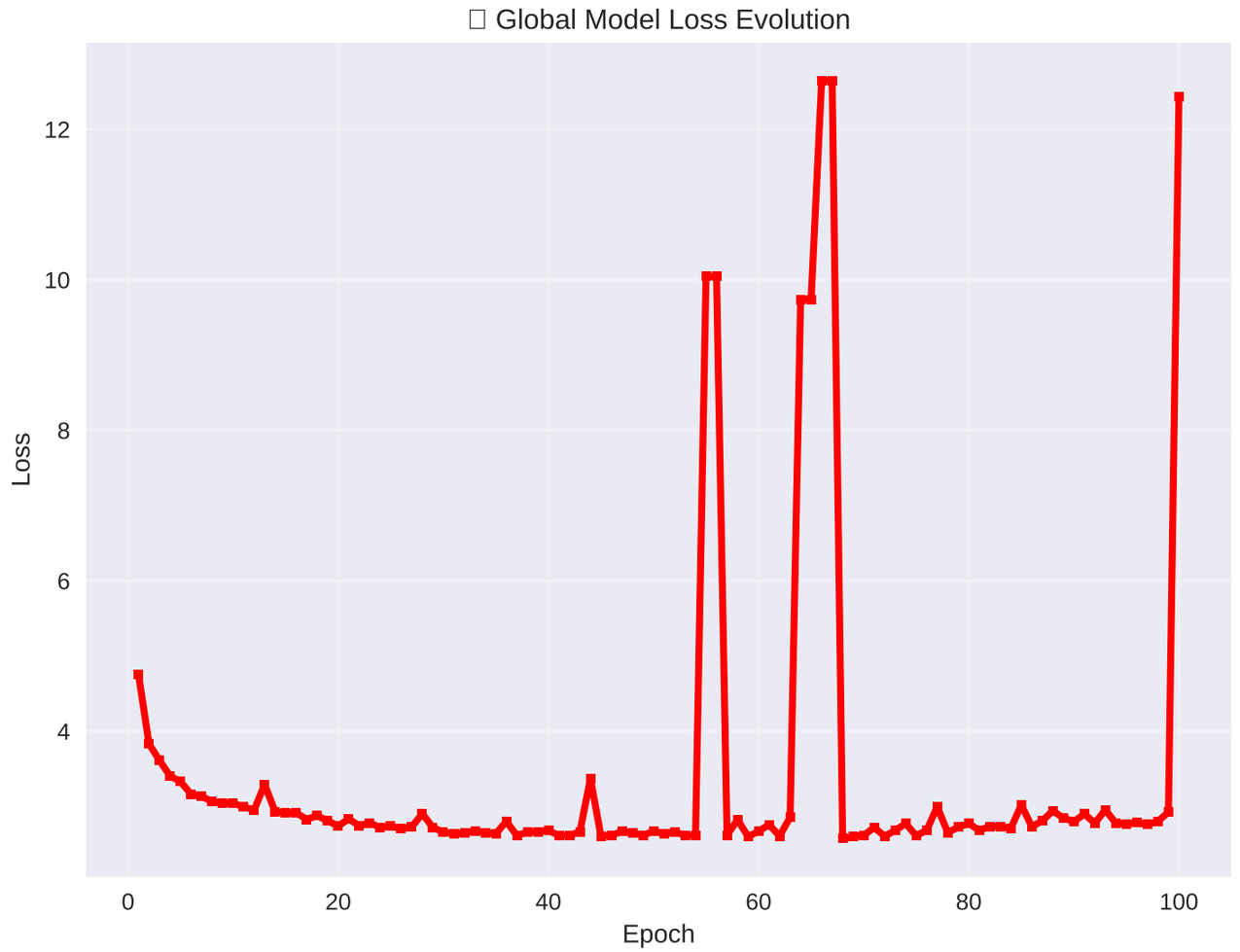
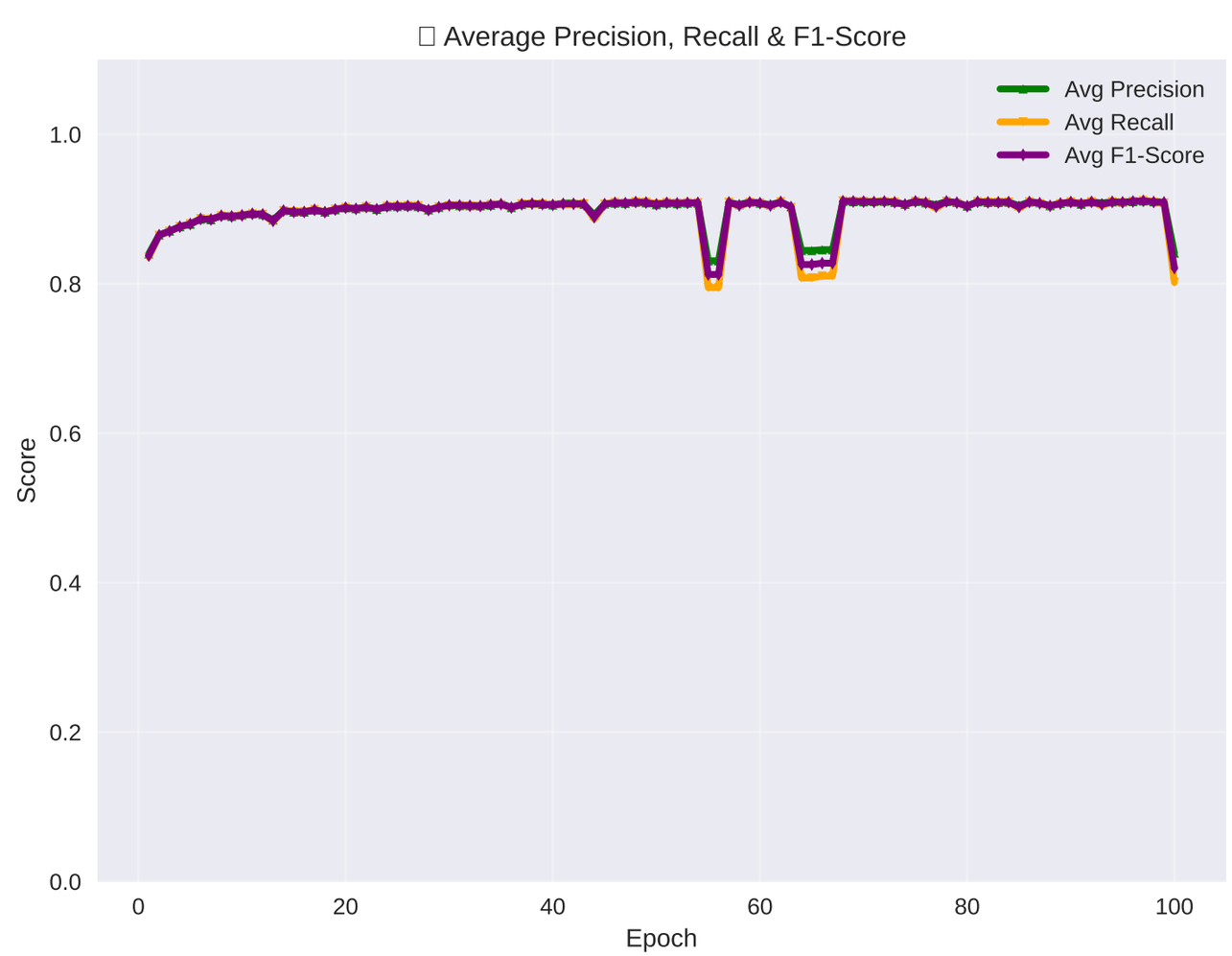
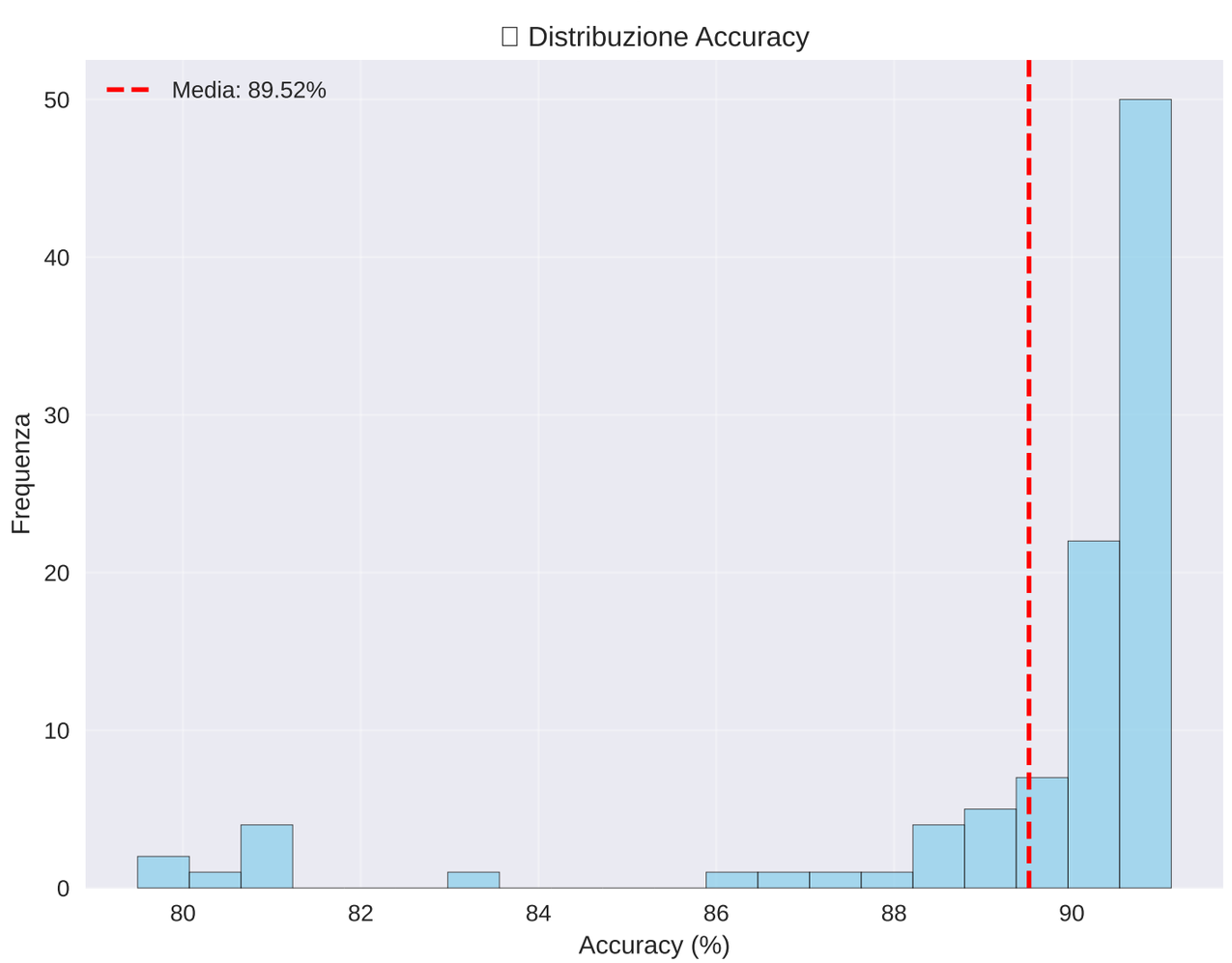
Federated Learning - Main Metrics Overview



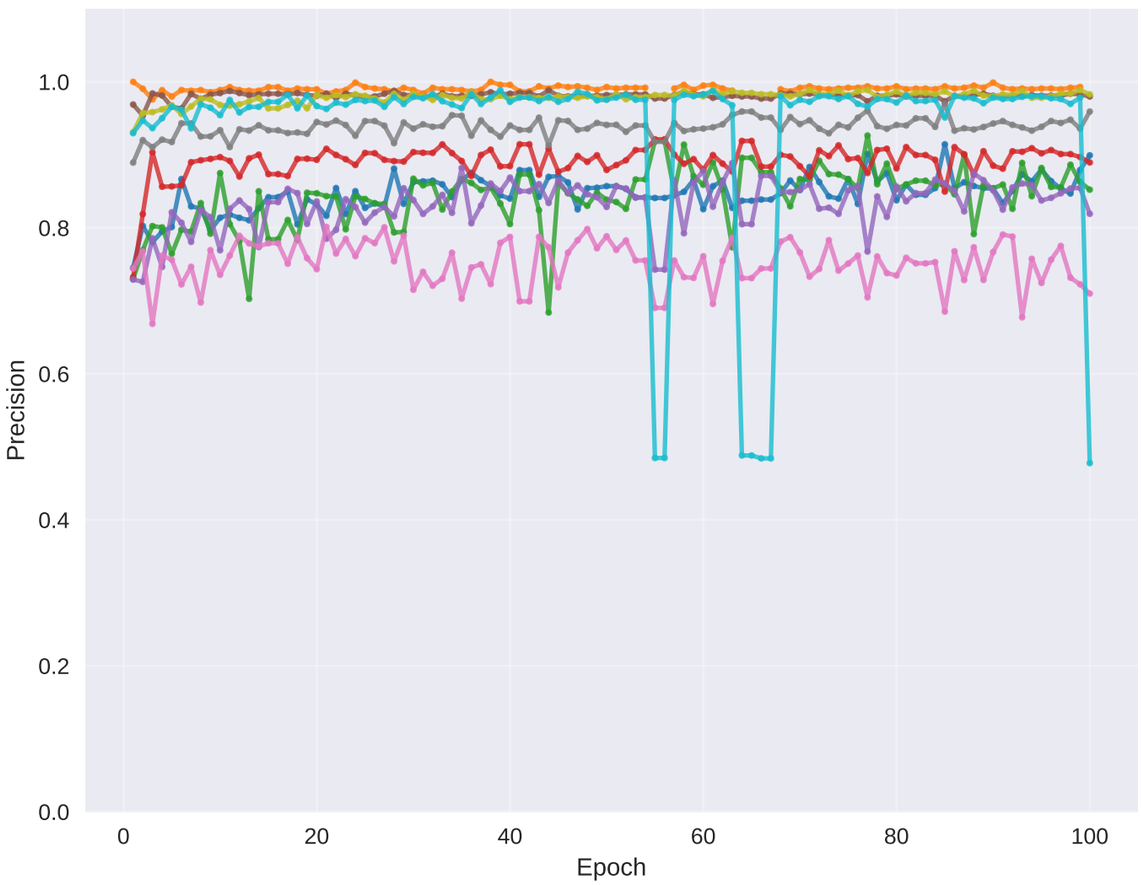




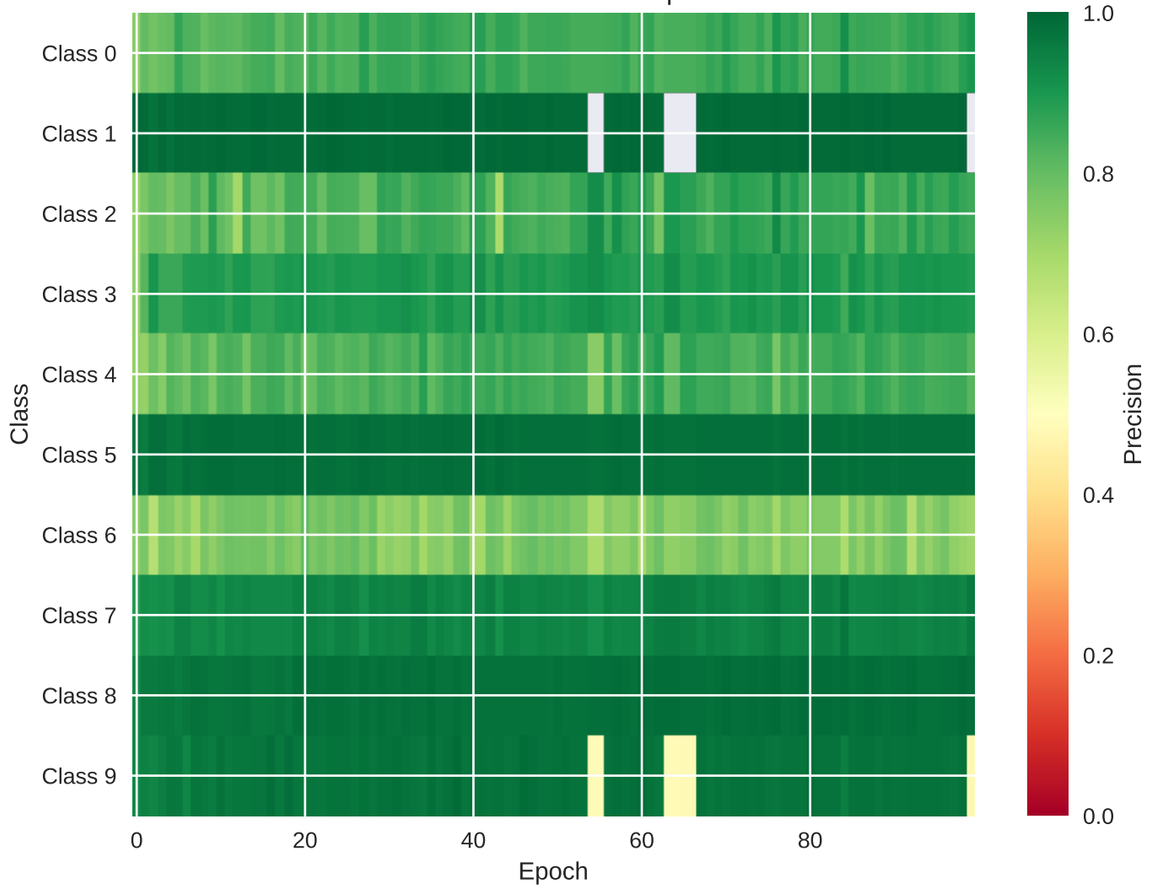


Per-Class Precision Analysis

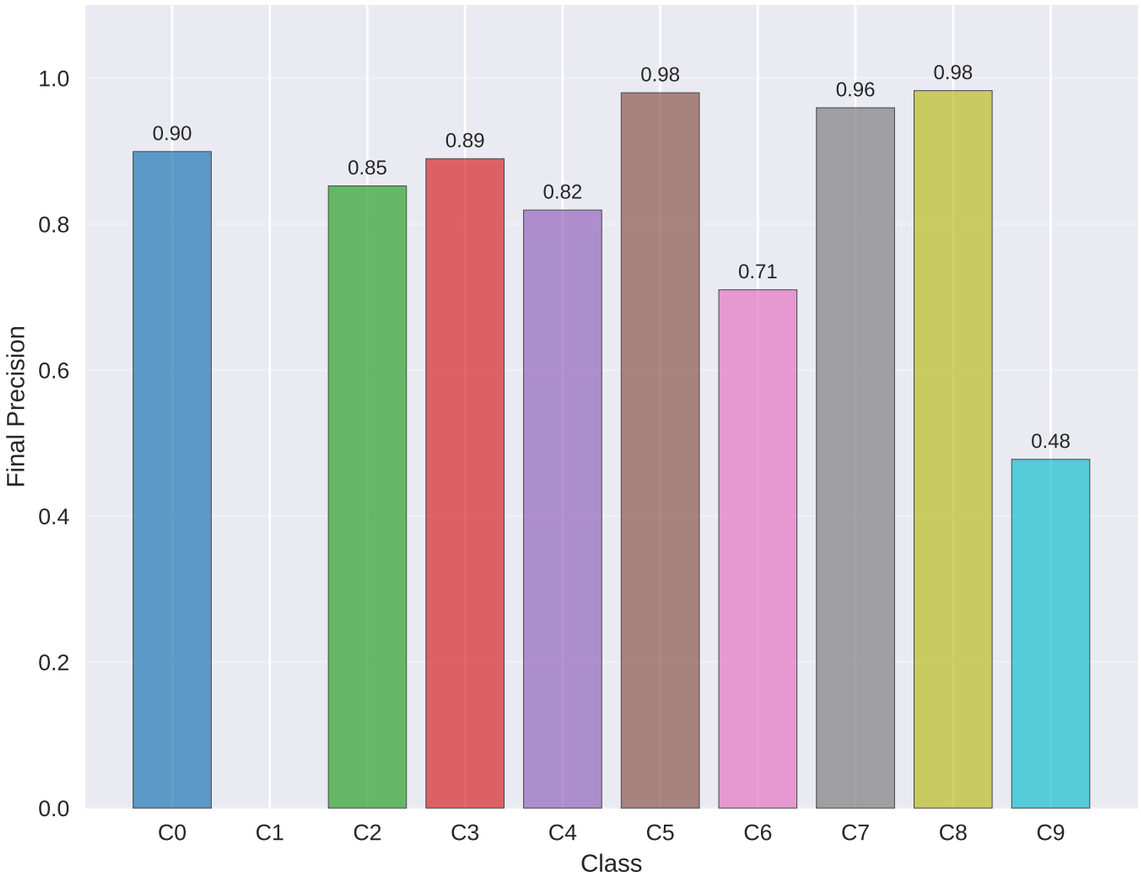
□ Per-Class Precision Evolution



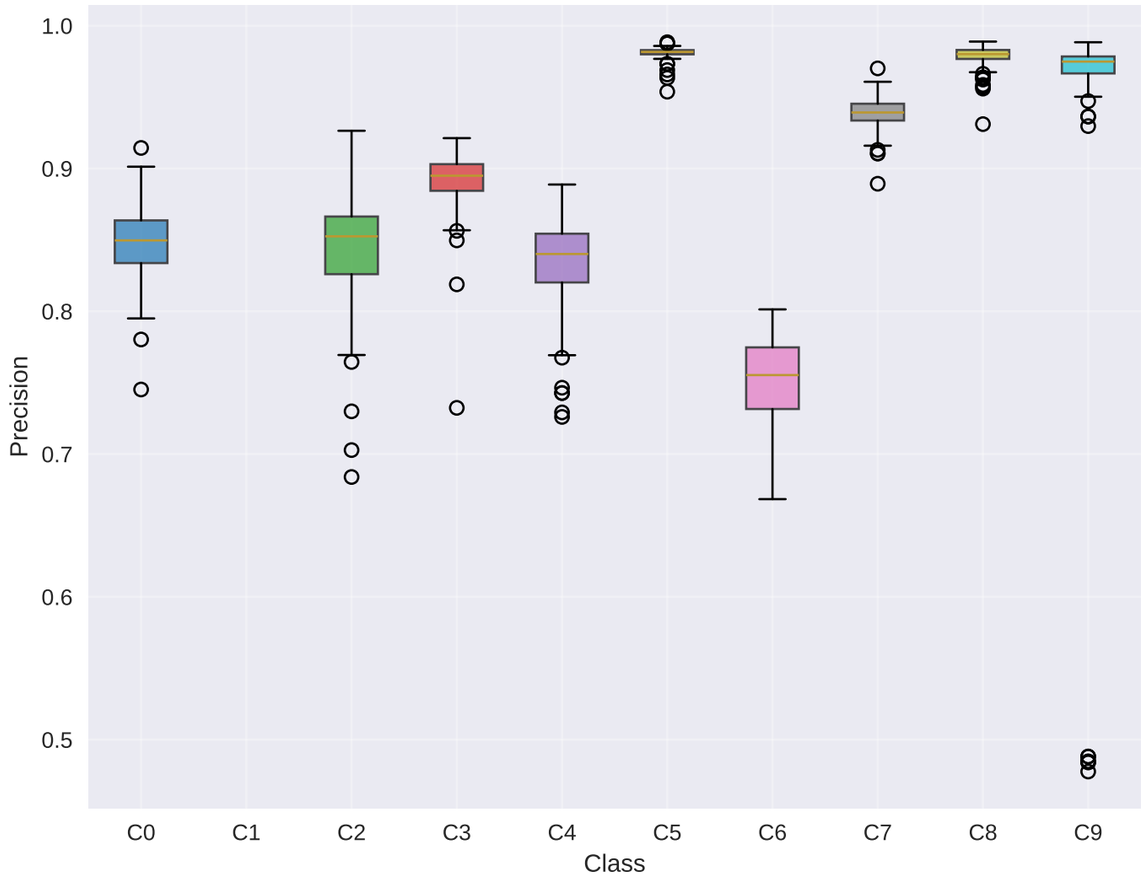
□ Precision Heatmap



□ Final Precision per Class

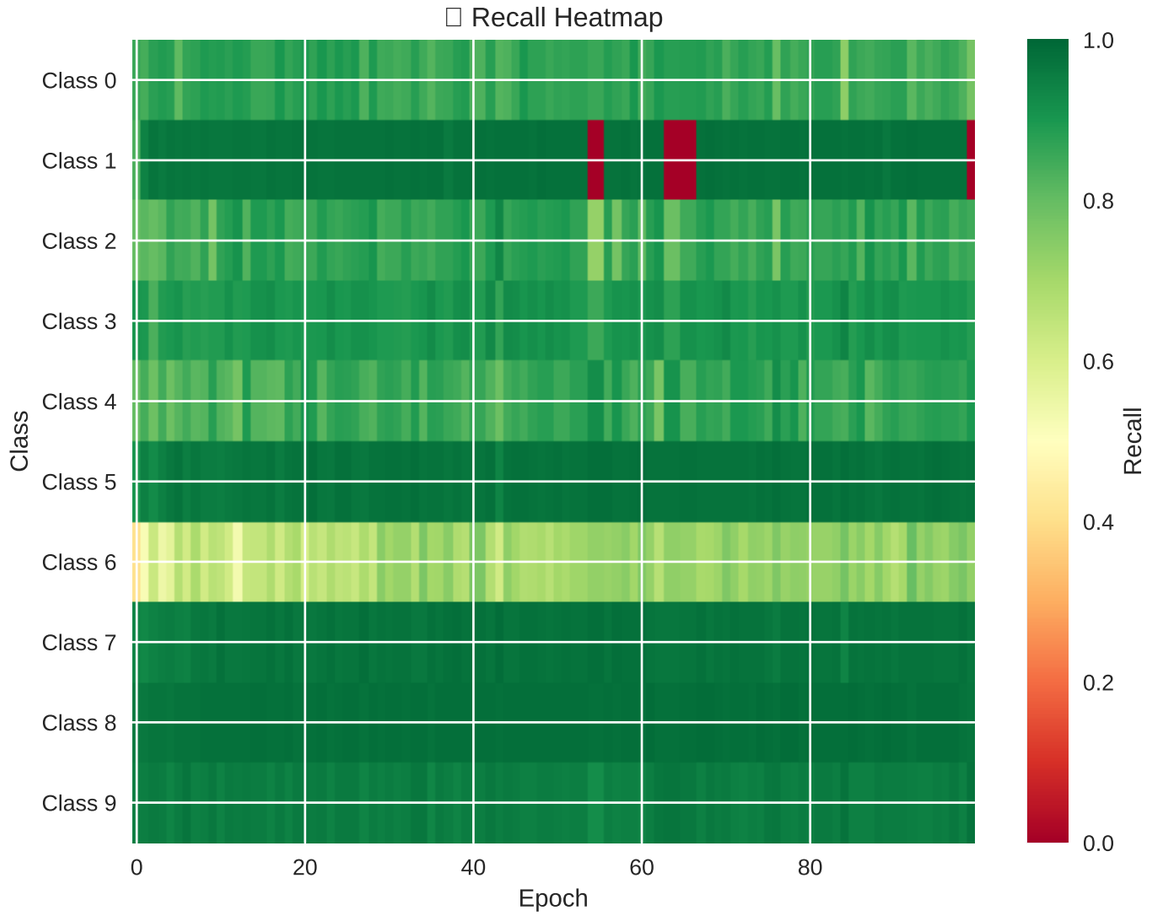


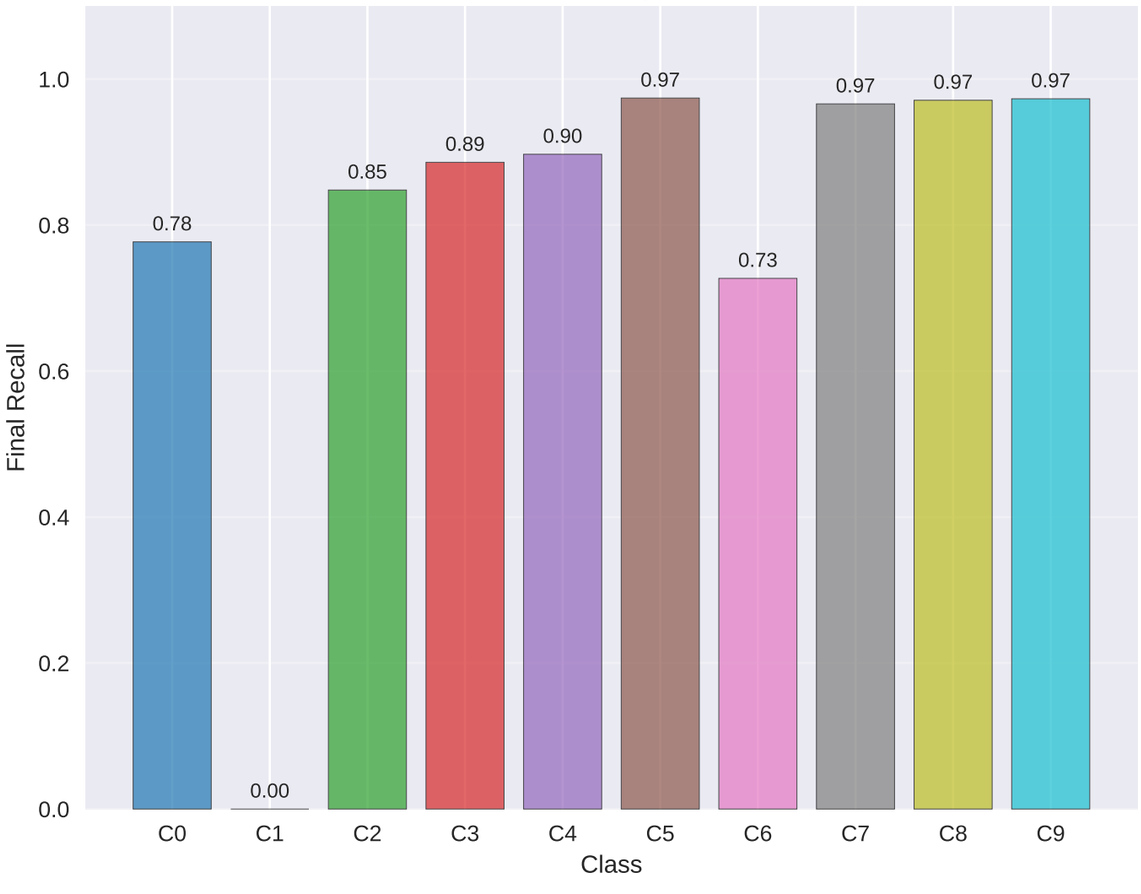
□ Precision Distribution per Class

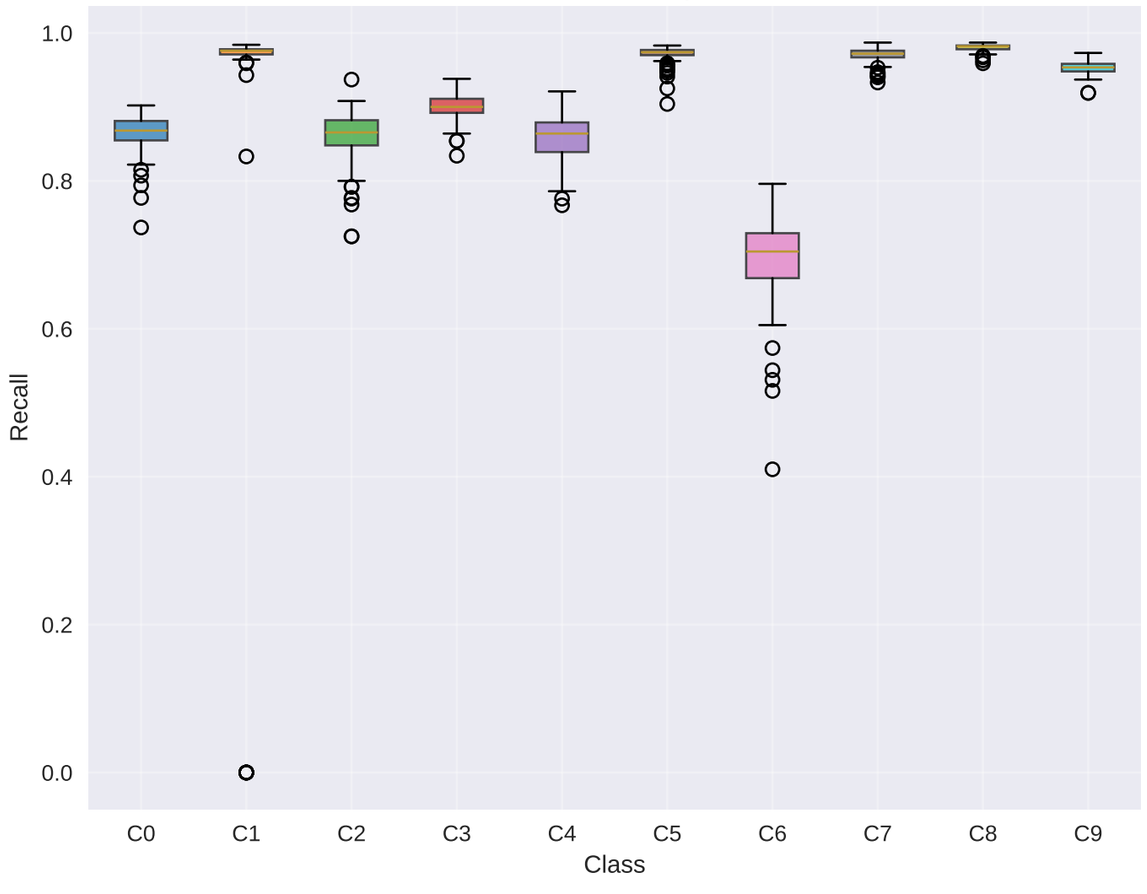


Per-Class Recall Analysis

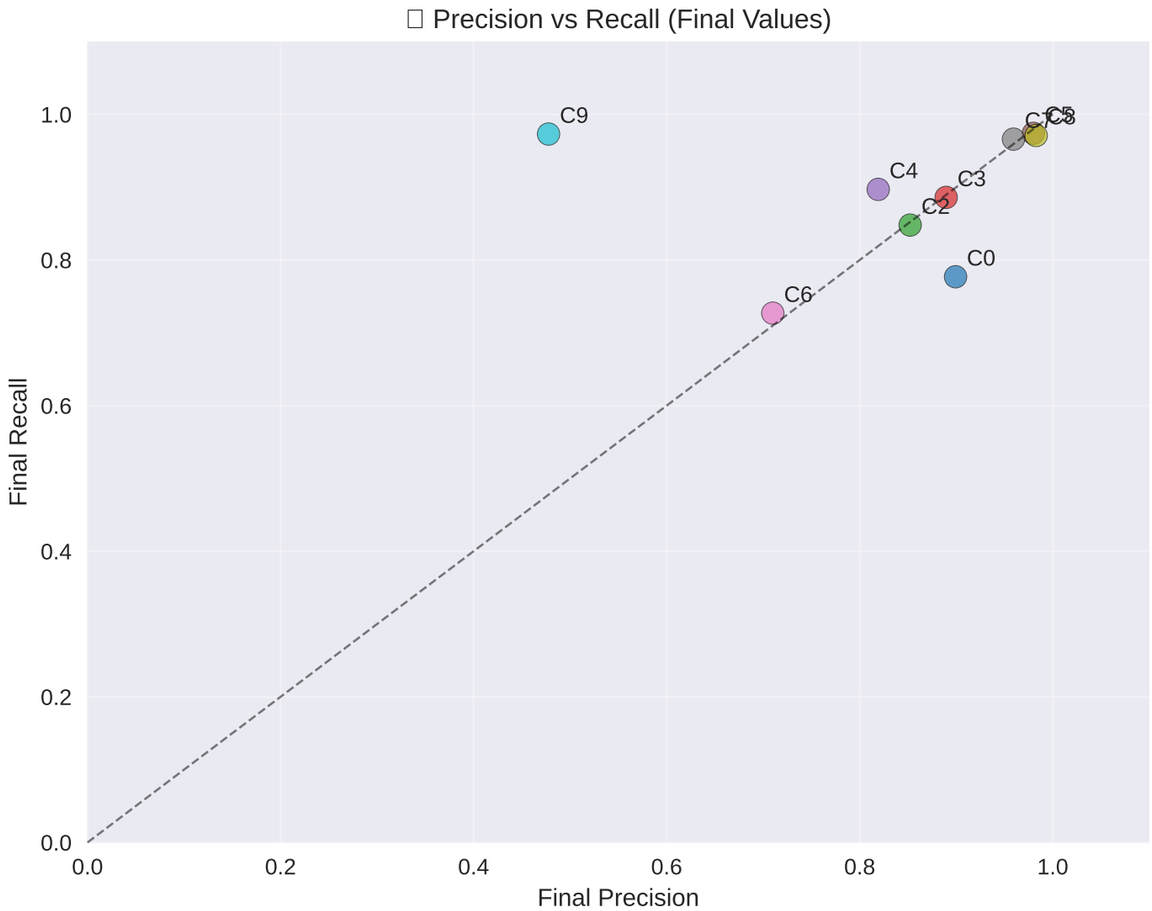


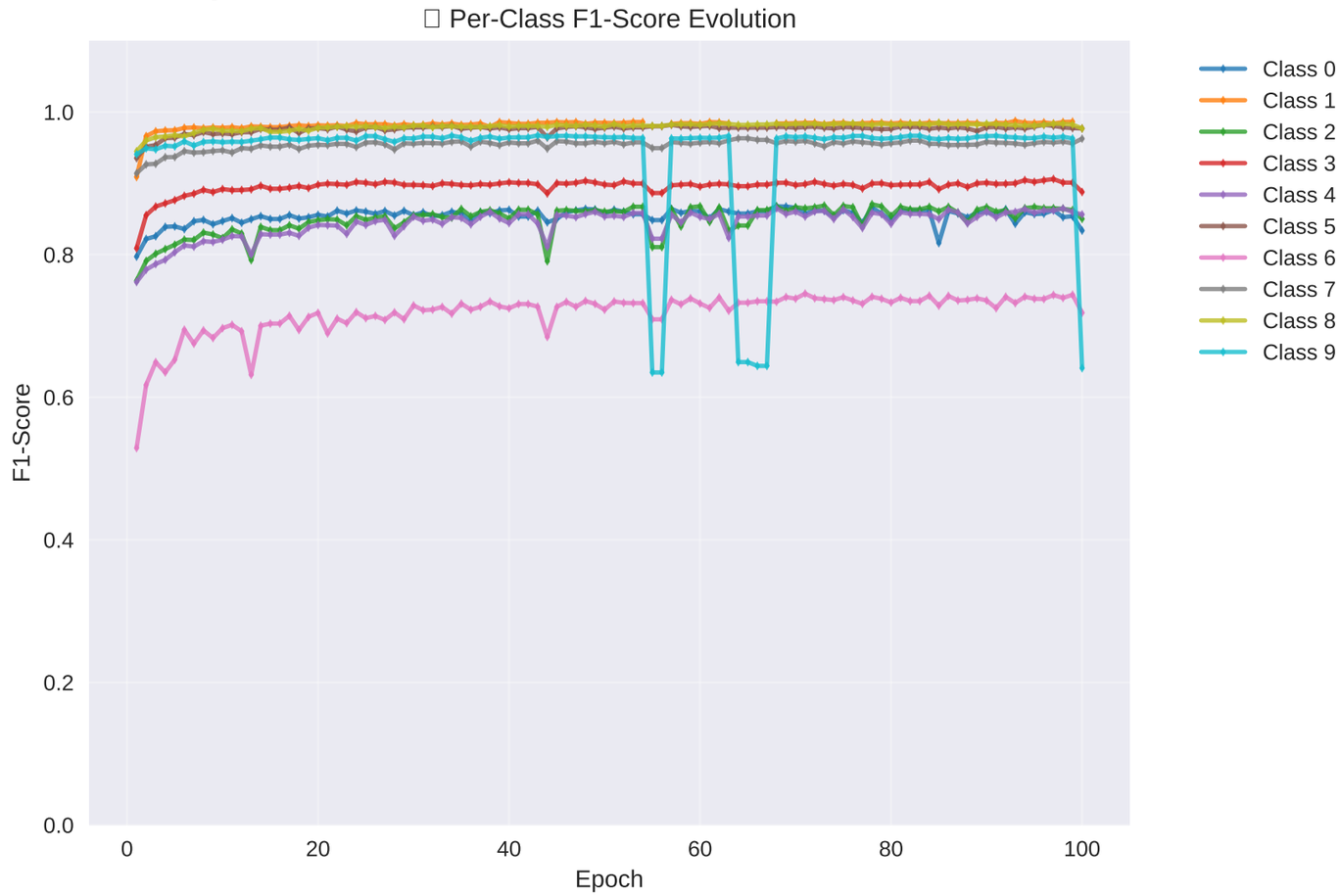


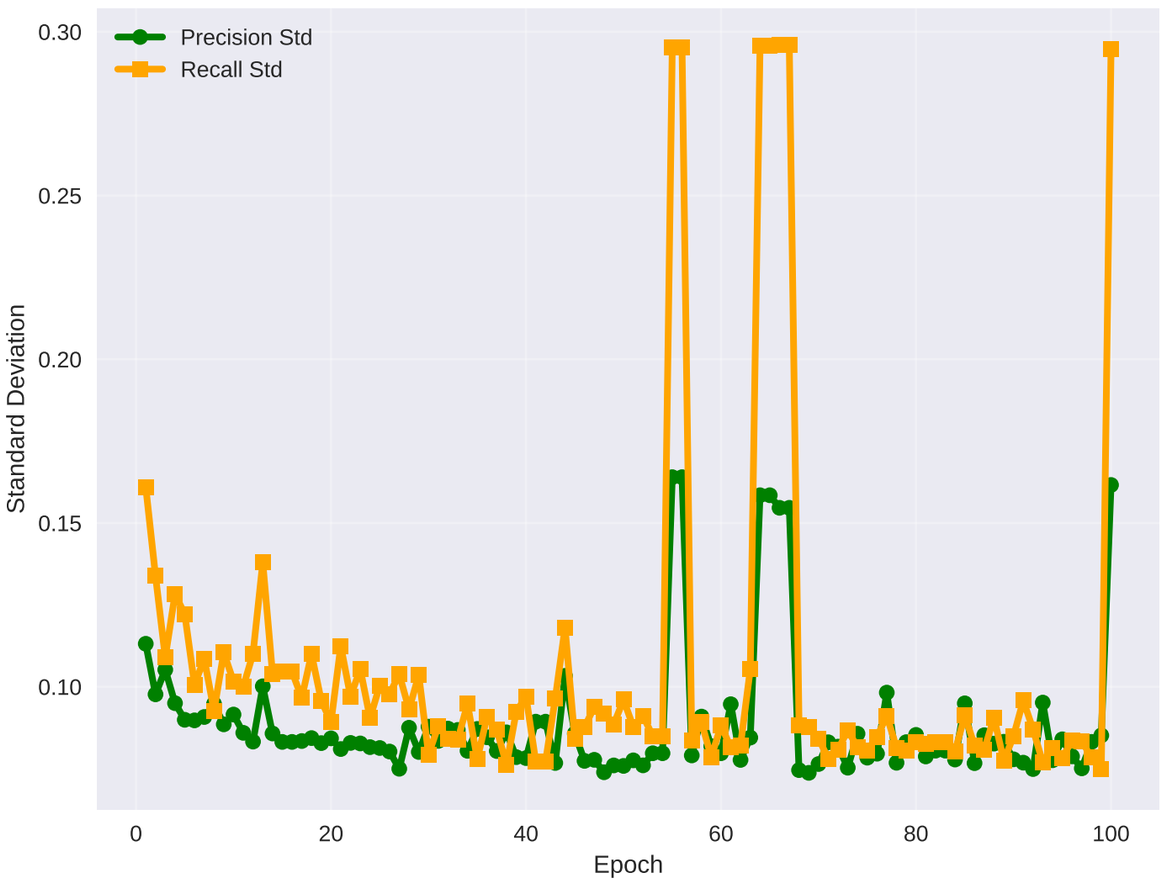


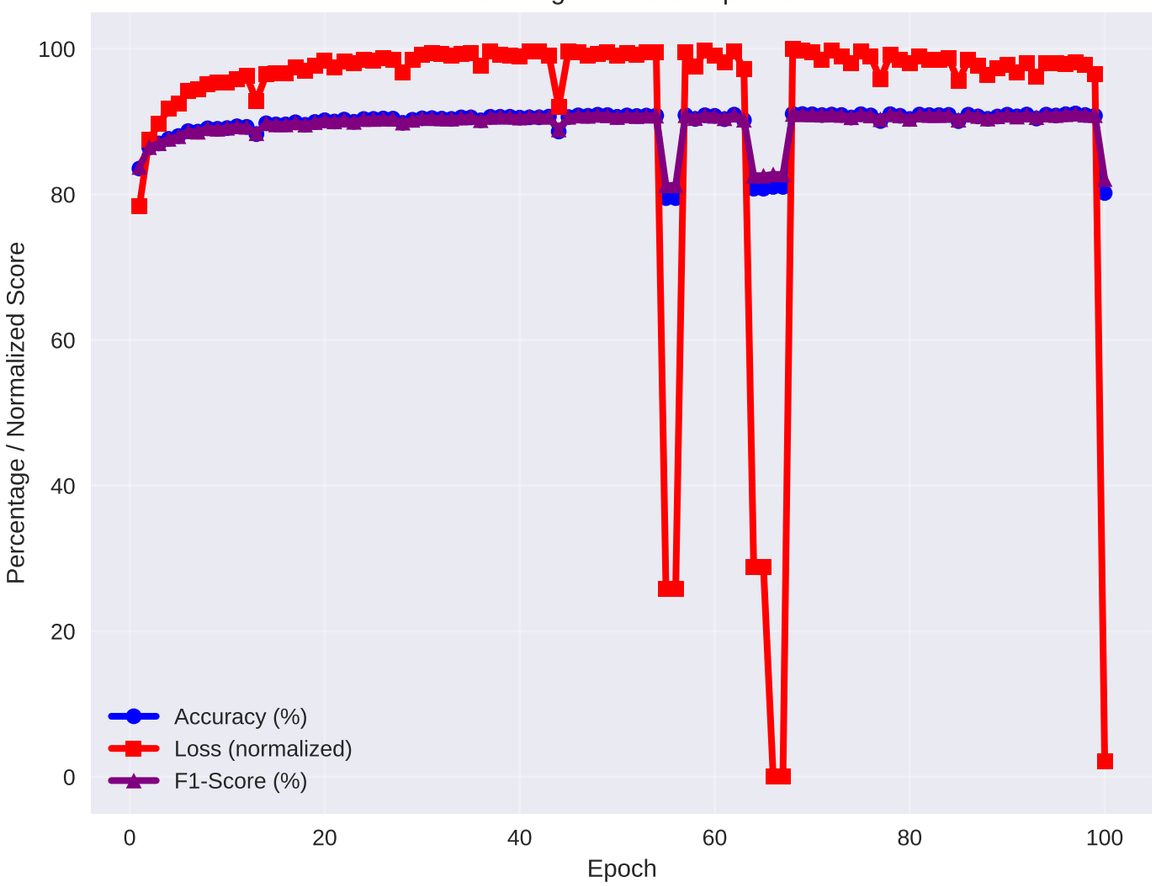


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-05 18:55:24
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 80.19%
• Loss: 12.4311
• Average Precision: 0.841
• Average Recall: 0.802
• Average F1-Score: 0.821

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: -3.36%
• Riduzione Loss: -7.6800
• Stabilità Accuracy (CV): 0.0307

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (nan)
• Peggiore Precision: Classe 1 (nan)
• Migliore Recall: Classe 5 (0.974)
• Peggiore Recall: Classe 1 (0.000)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.899, R=0.777, F1=0.834
• Classe 1: P=nan, R=0.000, F1=nan
• Classe 2: P=0.852, R=0.848, F1=0.850
• Classe 3: P=0.890, R=0.886, F1=0.888
• Classe 4: P=0.819, R=0.897, F1=0.856
• Classe 5: P=0.980, R=0.974, F1=0.977
• Classe 6: P=0.710, R=0.727, F1=0.718
• Classe 7: P=0.959, R=0.966, F1=0.963
• Classe 8: P=0.983, R=0.971, F1=0.977
• Classe 9: P=0.478, R=0.973, F1=0.641

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```